

# Krish Jeswal

Full-Stack Developer & ML Scientist | Electronics & Telecom Undergrad  
[connectwithkjeswal@gmail.com](mailto:connectwithkjeswal@gmail.com) | [918750208208](tel:918750208208) | <https://krish-jeswal.web.app> | <https://github.com/KrishJeswal>

## Summary

Undergrad researcher and full-stack engineer targeting IEEE ISCAS 2026 for ML side-channel attack work on masked AES-128. Builds and ships production AI systems — currently a RAG pipeline on GCP Cloud Run using Qdrant, LangChain, and Gemini API. Stack spans PyTorch, FastAPI, React/TypeScript, and embedded systems (ESP32, CAN bus).

## Education

|                                                |                                |
|------------------------------------------------|--------------------------------|
| <b>R.V. College of Engineering</b>             | Undergraduate                  |
| Electronics and Telecommunication              | Bengaluru, India • 2024 - 2028 |
| <b>Amity International School</b>              | Higher Secondary               |
| Physics, Chemistry, Maths and Computer Science | Ghaziabad, India • 2009 - 2023 |

## Skills

|                                                                                          |                                                                                |                                                            |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Programming Languages</b>                                                             | <b>AI/ML</b>                                                                   | <b>Data &amp; Viz</b>                                      |
| Python, C, JavaScript, TypeScript, SQL, Assembly, Verilog                                | PyTorch, scikit-learn, RAG, LangChain, Hugging Face, XGBoost, Gemini API, SHAP | NumPy, Pandas, OpenCV, Streamlit, Vertex AI, GCP           |
| <b>Web &amp; Backend</b>                                                                 | <b>Databases &amp; Tools</b>                                                   | <b>Electronics</b>                                         |
| React, Tailwind CSS, Vite, Three.js, Node.js, Express, FastAPI, Flask, REST APIs, Docker | PostgreSQL, MongoDB, Qdrant, Supabase, Firebase, Git, CI/CD                    | Arduino, STM32, ESP32, FPGA, UART, LTSpice, KiCAD, CAN Bus |

## Projects

### CipherTrace

- Submitted to IEEE Access — profiling attack on masked AES-128 (ASCAD dataset, 50k traces) evaluating 72 classifier configurations across 6 models × 3 POI strategies; best result achieves Guessing Entropy of 16.36 at 500 traces using MLP + PCA.
- Demonstrated Identity labels (256-class) reduce GE to 0.46 vs. 12.03 for Hamming Weight, overturning a standard SCA assumption; SHAP-based leakage localization revealed only 20% overlap between model-identified and SNR-selected regions.

### Code Noir

- Built a RAG pipeline that transforms GitHub commit history into genre fiction (noir, sci-fi) for developer explainability — embeds commits with BGE-large-en-v1.5 via Hugging Face, stores in Qdrant, and retrieves at sub-100ms latency.
- Orchestrated Gemini generation via LangChain prompt pipeline; deployed FastAPI backend on GCP Cloud Run with Docker, React/TypeScript frontend, and Firebase auth — full production system on a serverless architecture.

### Intelligent Excel Parser

- Built a FastAPI + Gemini (gemini-2.5-flash-lite) service that eliminates manual column remapping for enterprise .xlsx ingestion — achieved 94% mapping accuracy across 300+ header variants with automatic SequenceMatcher fallback.
- Engineered modular pipeline with Pydantic v2 schema validation and openpyxl; surfaced per-cell confidence scores via an interactive dashboard, cutting estimated data-cleaning overhead by 80%.

## Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                      |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Google Developer Groups - Research &amp; Projects</b>                                                                                                                                                                                                                                                                                                                                                             | Jun 2025 – Present |
| Designed two end-to-end GCP projects (Vertex AI fine-tuning, Cloud Run deployment) as part of the Google Cloud Study Jams; provided technical mentorship to 15+ teams at Vision 2047 Hackathon on ML pipeline architecture and cloud deployment; hosted a DevSprint session on open source contribution and cybersecurity fundamentals, guiding participants through hands-on workflows and security best practices. |                    |
| <b>The Institution of Electronics and Telecommunication Engineers (IETE) - Development Lead</b>                                                                                                                                                                                                                                                                                                                      | Nov 2025 – Present |
| Led the development team to architect and ship the official IETE Student Forum website — defined the full stack (React, TypeScript, Tailwind, PostgreSQL), established CI/CD via GitHub Actions, and drove end-to-end delivery from system design to production deployment. Introduced code review workflows and task delegation structure adopted across the chapter.                                               |                    |